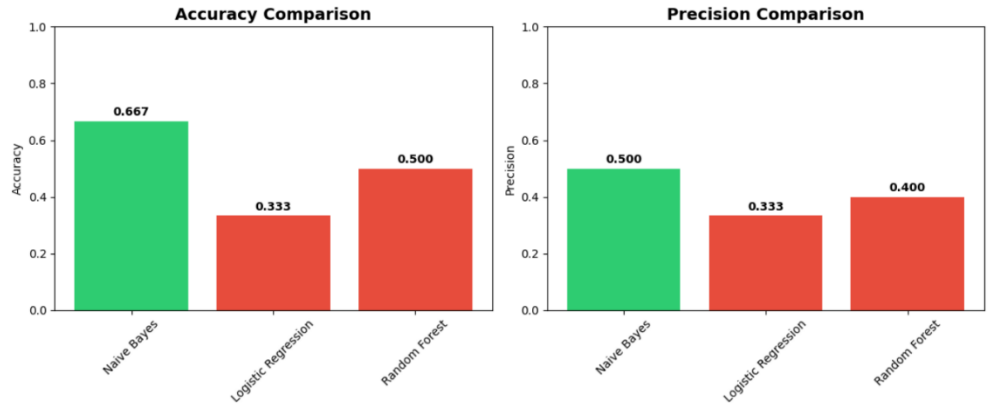
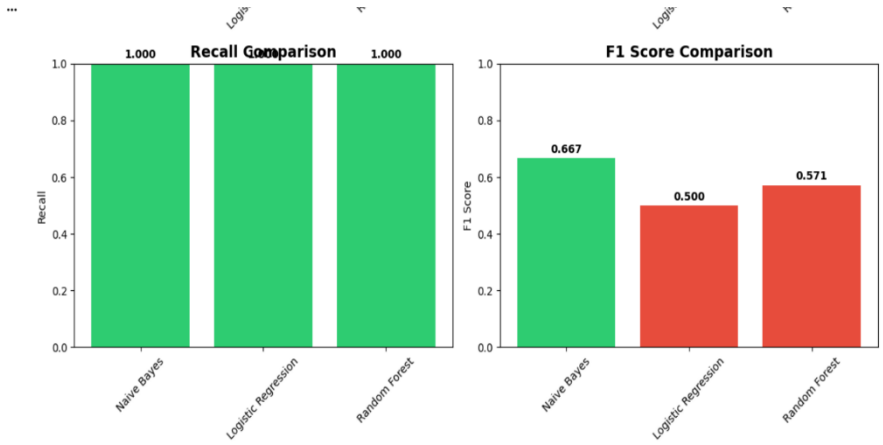
OUTPUTS

Google Collab outputs:

```
🏆 Best Performing Model: Naive Bayes
🎨 Generating visualizations...
```






```
ui.create_ui()

print("\n✅ System Ready! Use the interface above to detect fake news.")
```

```
🚀 Starting Fake News Detection System...
============================================================

📊 Creating training dataset...
✅ Created dataset with 30 articles (15 Real, 15 Fake)

🔧 Preprocessing text data...
✅ Preprocessing complete

🤖 Initializing ML models...

📈 Extracting TF-IDF features...
✅ Training samples: 24, Testing samples: 6

🎯 Training ML models...
Training Naive Bayes...
Training Logistic Regression...
Training Random Forest...

📊 Evaluating model performance...

============================================================
📊 MODEL EVALUATION RESULTS
============================================================
                     Accuracy  Precision  Recall  F1 Score
Naive Bayes            0.6667     0.5000     1.0    0.6667
```

Fake_News_Detection_System.ipynb Save in GitHub to keep changes

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all Copy to Drive

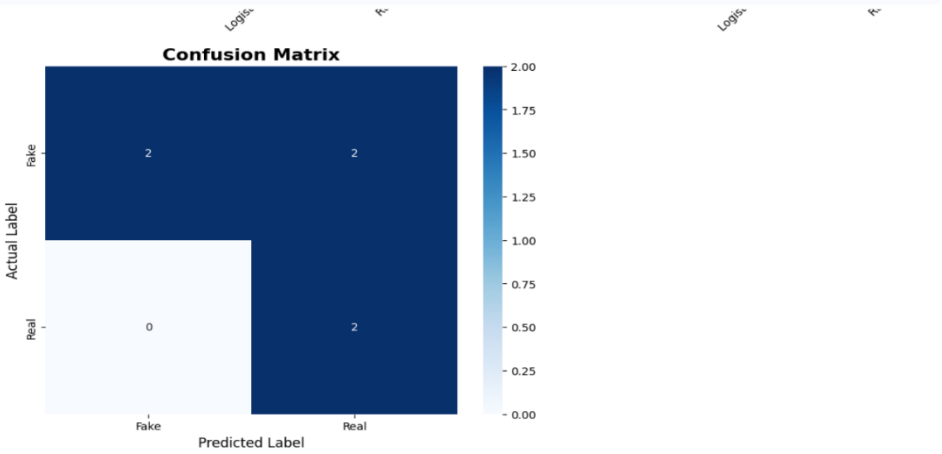
```
storage = DataStorage()
ui = FakeNewsUI(detector, preprocessor, storage)
ui.create_ui()

print("\n✅ System Ready! Use the interface above to detect fake news.")
```

```
🎯 Training ML models...
Training Naive Bayes...
Training Logistic Regression...
Training Random Forest...

📊 Evaluating model performance...

============================================================
📊 MODEL EVALUATION RESULTS
============================================================
                      Accuracy  Precision  Recall  F1 Score
Naive Bayes             0.6667     0.5000     1.0    0.6667
Logistic Regression     0.3333     0.3333     1.0    0.5000
Random Forest           0.5000     0.4000     1.0    0.5714

🏆 Best Performing Model: Naive Bayes

🎨 Generating visualizations...
```

Fake Real

Predicted Label

```
============================================================
📈 Naive Bayes - Final Performance Metrics
============================================================
Accuracy:  0.6667
Precision: 0.5000
Recall:    1.0000
F1 Score:  0.6667

🎨 Launching Interactive User Interface...
```



Commands + Code + Text Run all Copy to Drive

```
🎨 Launching Interactive User Interface...
```

Fake News Detection System

Powered by Machine Learning & NLP | **No MySQL Required**

Predict News | Dashboard | History

Enter News Article Text:

```
SHOCKING: The government has secretly admitted that aliens exist and have been living among us for decades. According to anonymous sources inside the White House, a secret agreement was signed with extraterrestrials in 1947. The mainstream media is hiding this truth from the public!
```

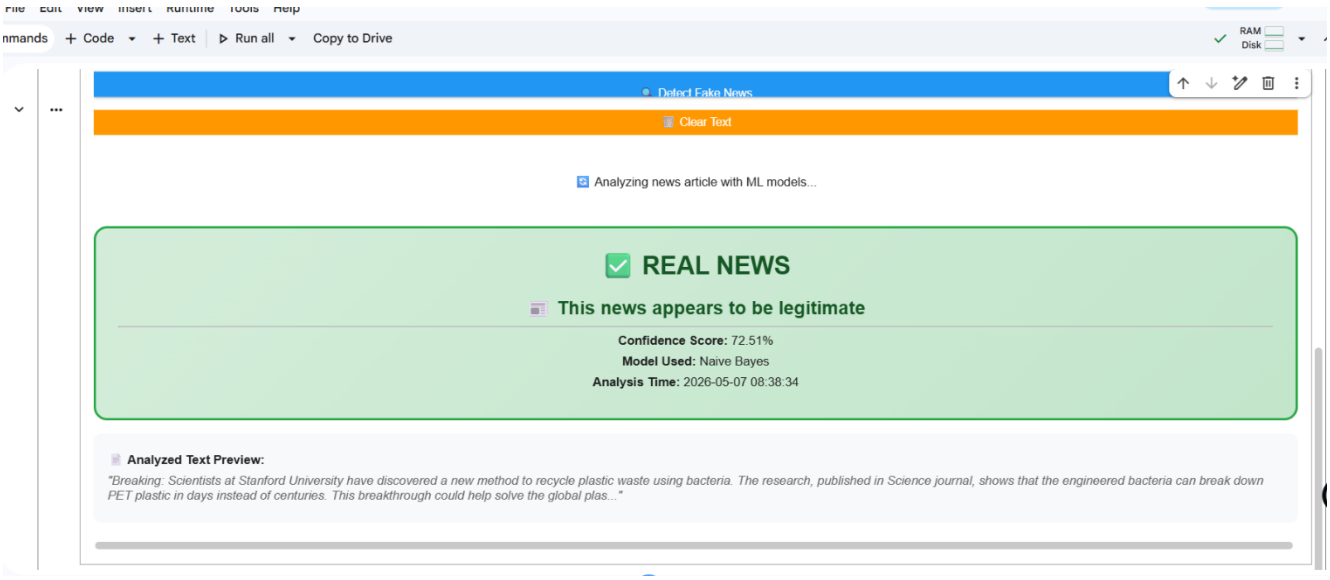
Try Examples:

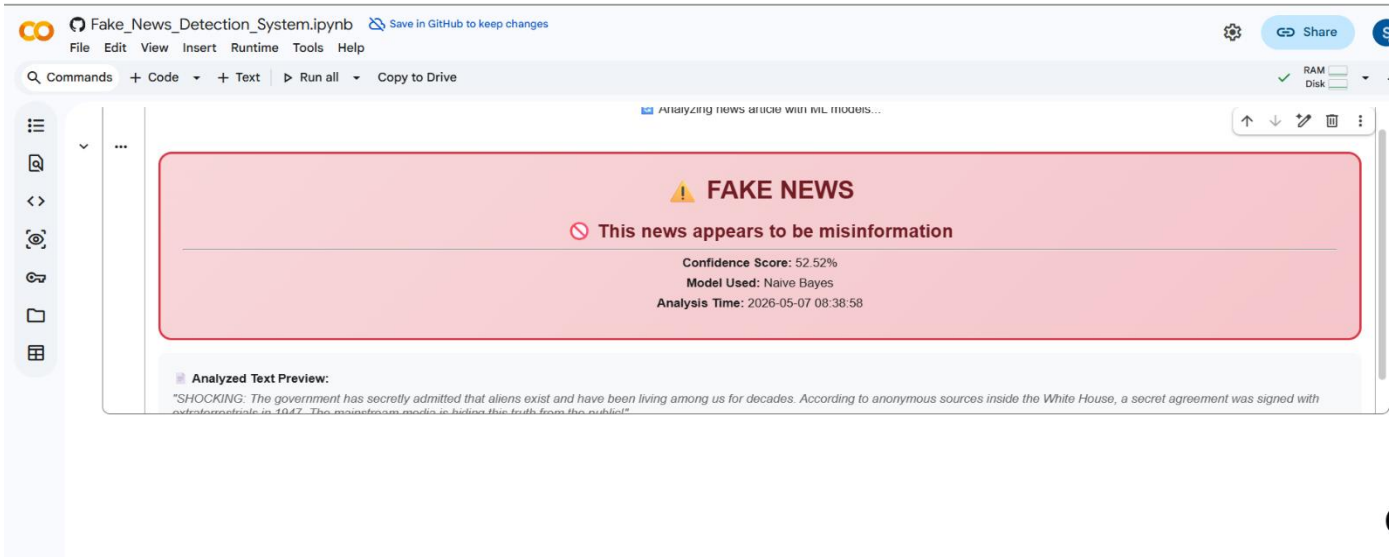
Real News Example | Fake News Example

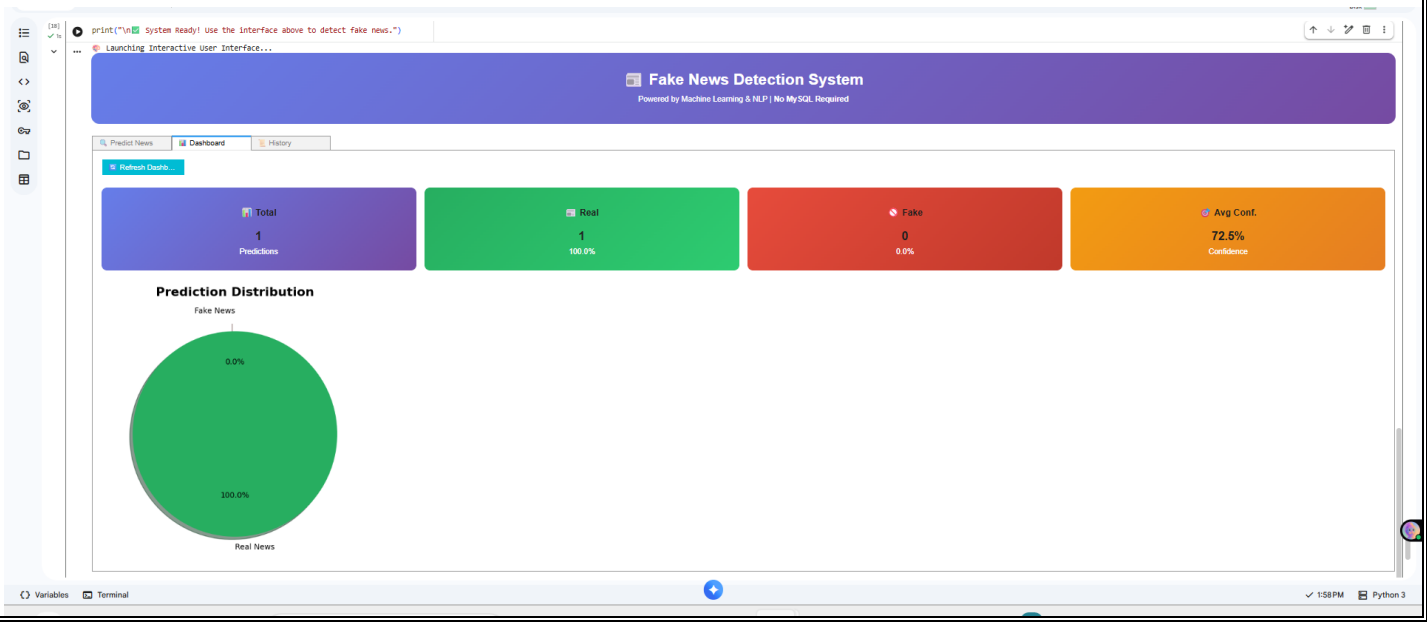
Detect Fake News

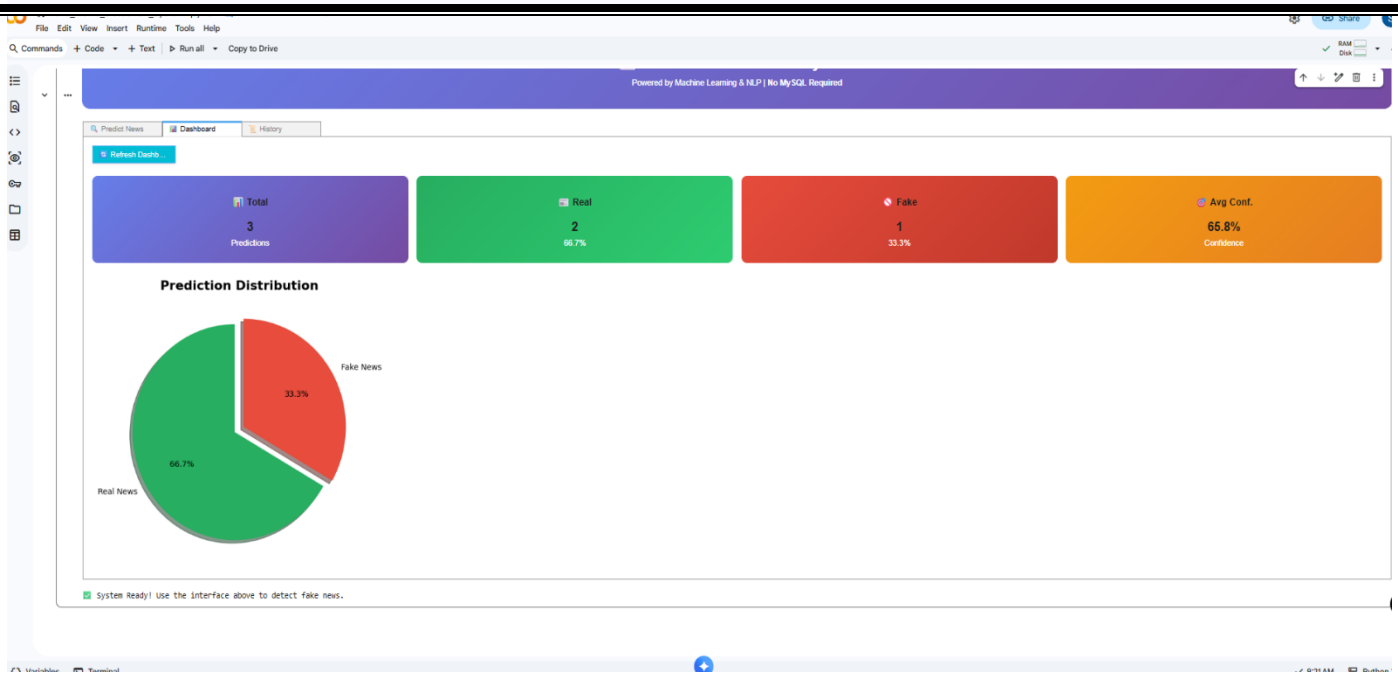
Clear Text

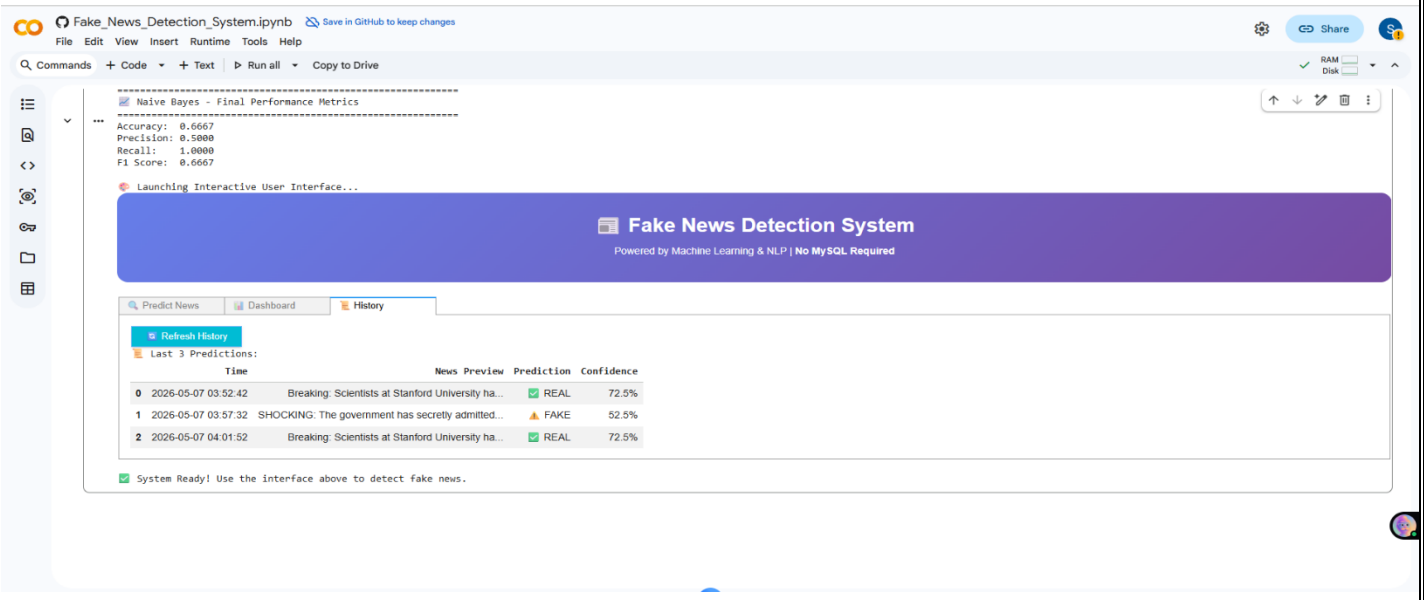
```
🔍 Analyzing news article with ML models...
```


File Edit View Insert Runtime Tools Help
Code
Text
Run all
Copy to Drive
RAM
Disk
Detect Fake News
Clear Text
Analyzing news article with ML models...
REAL NEWS
This news appears to be legitimate
Confidence Score: 72.51%
Model Used: Naive Bayes
Analysis Time: 2026-05-07 08:38:34
Analyzed Text Preview:
"Breaking: Scientists at Stanford University have discovered a new method to recycle plastic waste using bacteria. The research, published in Science journal, shows that the engineered bacteria can break down PET plastic in days instead of centuries. This breakthrough could help solve the global plas..."


Fake_News_Detection_System.ipynb
Save in GitHub to keep changes
File Edit View Insert Runtime Tools Help
Share
Commands
Code
Text
Run all
Copy to Drive
RAM
Disk
FAKE NEWS
This news appears to be misinformation
Confidence Score: 52.52%
Model Used: Naive Bayes
Analysis Time: 2026-05-07 08:38:58
Analyzed Text Preview:
"SHOCKING: The government has secretly admitted that aliens exist and have been living among us for decades. According to anonymous sources inside the White House, a secret agreement was signed with


System Ready! Use the interface above to detect fake news.
Launching Interactive User Interface...
Fake News Detection System
Powered by Machine Learning & NLP | No MySQL Required
Predict News
Dashboard
History
Refresh Dashb...
Total
1
Predictions
Real
1
100.0%
Fake
0
0.0%
Avg Conf.
72.5%
Confidence
Prediction Distribution
Fake News
0.0%
100.0%
Real News
Variables
Terminal
1:58 PM
Python 3


File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all Copy to Drive
Powered by Machine Learning & NLP | No MySQL Required
Predict News
Dashboard
History
Refresh Dashb...
Total
3
Predictions
Real
2
66.7%
Fake
1
33.3%
Avg Conf.
65.8%
Confidence
Prediction Distribution
Fake News
33.3%
66.7%
Real News
System Ready! Use the interface above to detect fake news.


Fake_News_Detection_System.ipynb
Save in GitHub to keep changes
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all Copy to Drive
Share
Naive Bayes - Final Performance Metrics
Accuracy: 0.6667
Precision: 0.5000
Recall: 1.0000
F1 Score: 0.6667
Launching Interactive User Interface...
Fake News Detection System
Powered by Machine Learning & NLP | No MySQL Required
Predict News
Dashboard
History
Refresh History
Last 3 Predictions:
Time
News Preview
Prediction
Confidence
0 2026-05-07 03:52:42 Breaking: Scientists at Stanford University ha... REAL 72.5%
1 2026-05-07 03:57:32 SHOCKING: The government has secretly admitted... FAKE 52.5%
2 2026-05-07 04:01:52 Breaking: Scientists at Stanford University ha... REAL 72.5%
System Ready! Use the interface above to detect fake news.